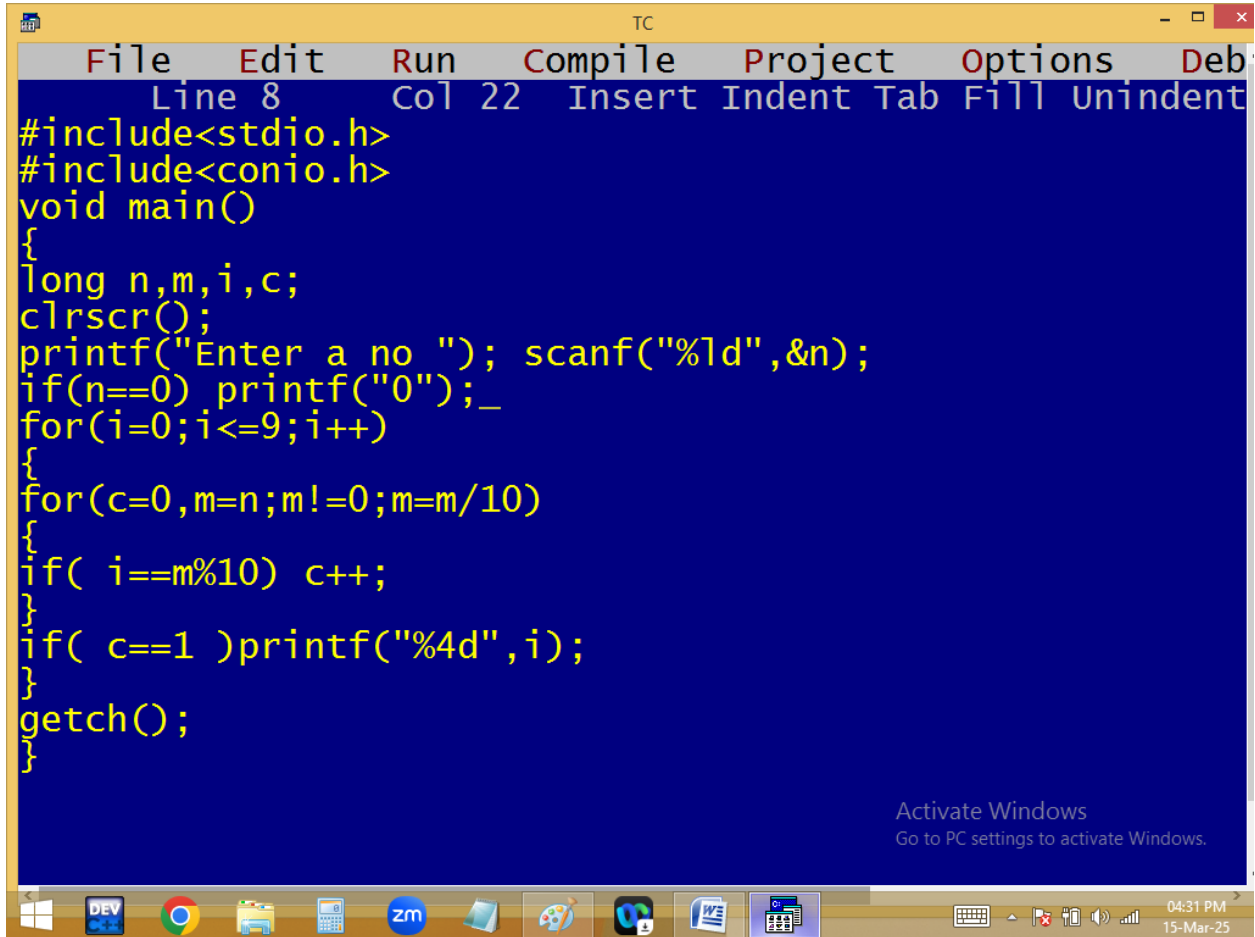


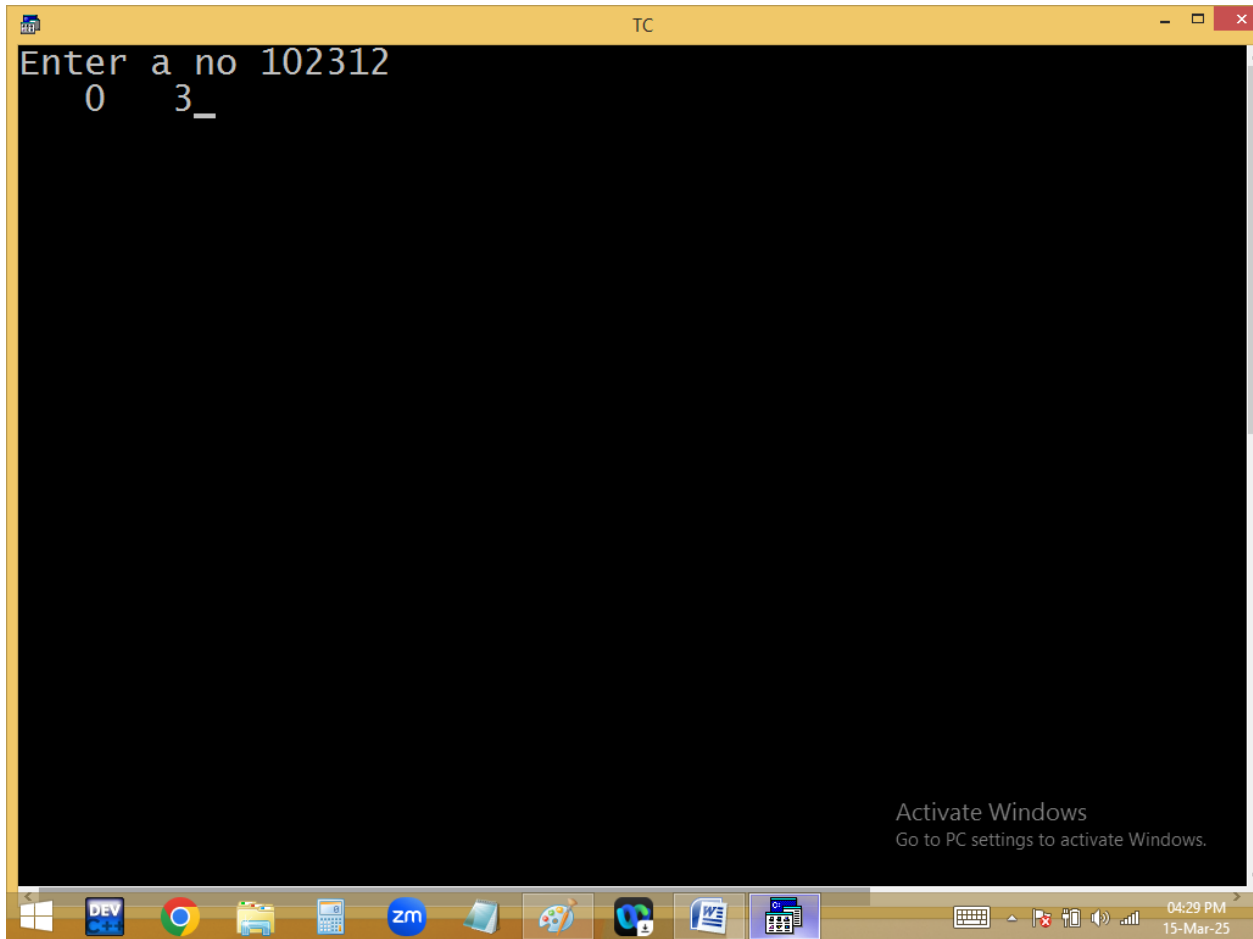
Finding unique digits in given no.

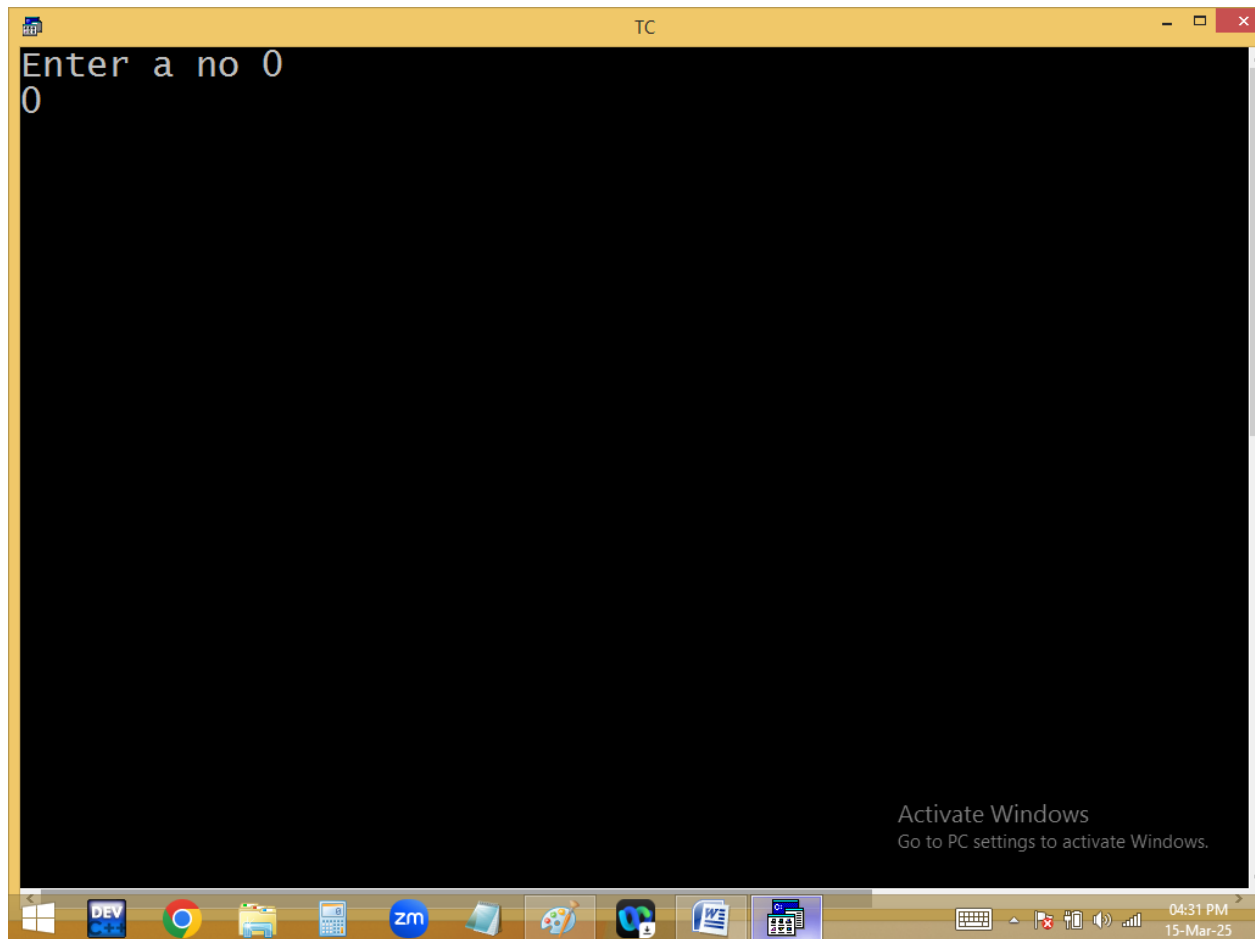


The screenshot shows a Turbo C++ (TC) IDE window with a blue background and yellow text. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom indicates 'Line 8 Col 22'. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,m,i,c;
    clrscr();
    printf("Enter a no "); scanf("%ld",&n);
    if(n==0) printf("0");
    for(i=0;i<=9;i++)
    {
        for(c=0,m=n;m!=0;m=m/10)
        {
            if( i==m%10) c++;
        }
        if( c==1 )printf("%4d",i);
    }
    getch();
}
```

An 'Activate Windows' watermark is visible in the bottom right corner of the IDE window. The Windows taskbar at the bottom shows various icons including DEV, Chrome, File Explorer, Calculator, Zoom, and others, along with the system clock showing 04:31 PM on 15-Mar-25.



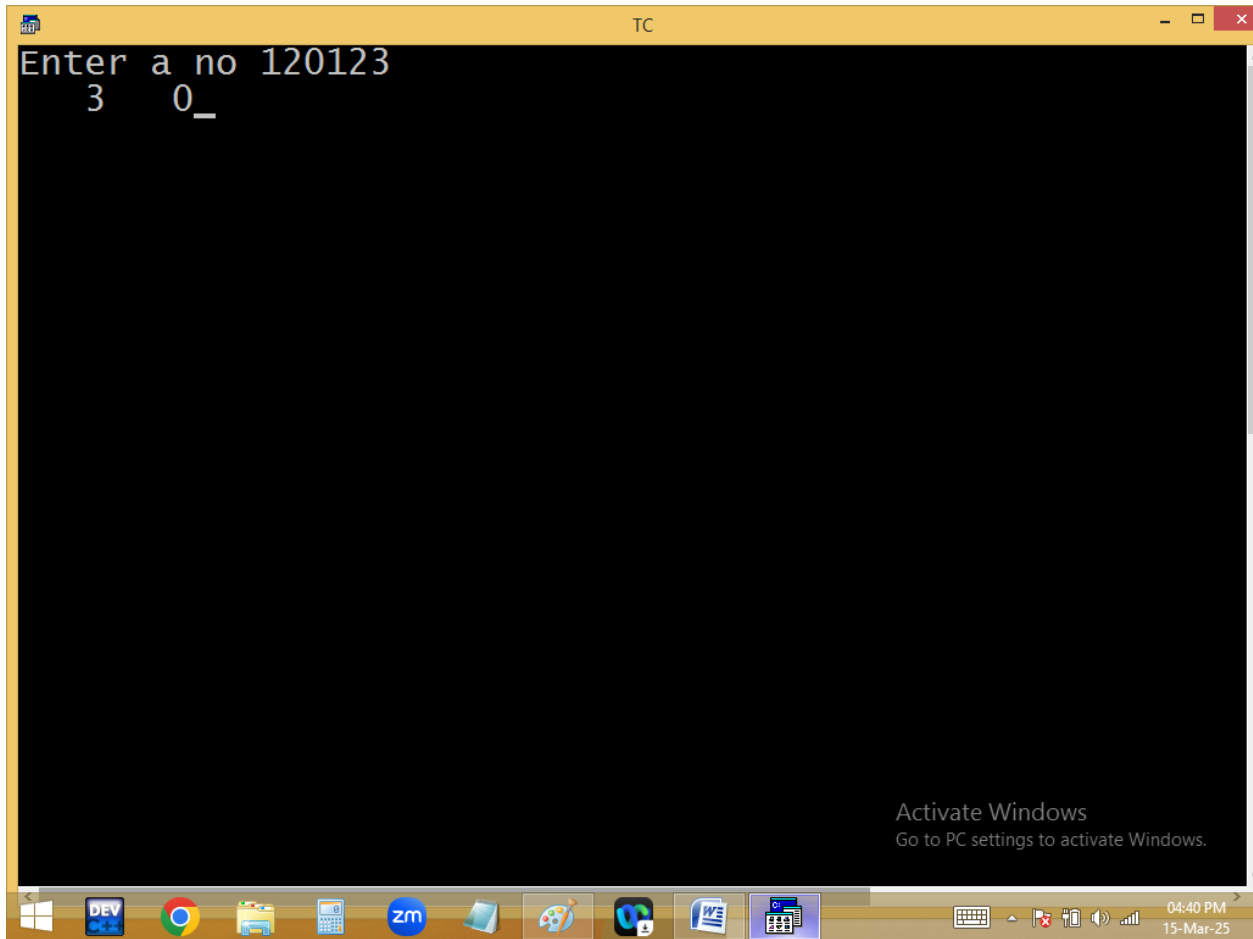


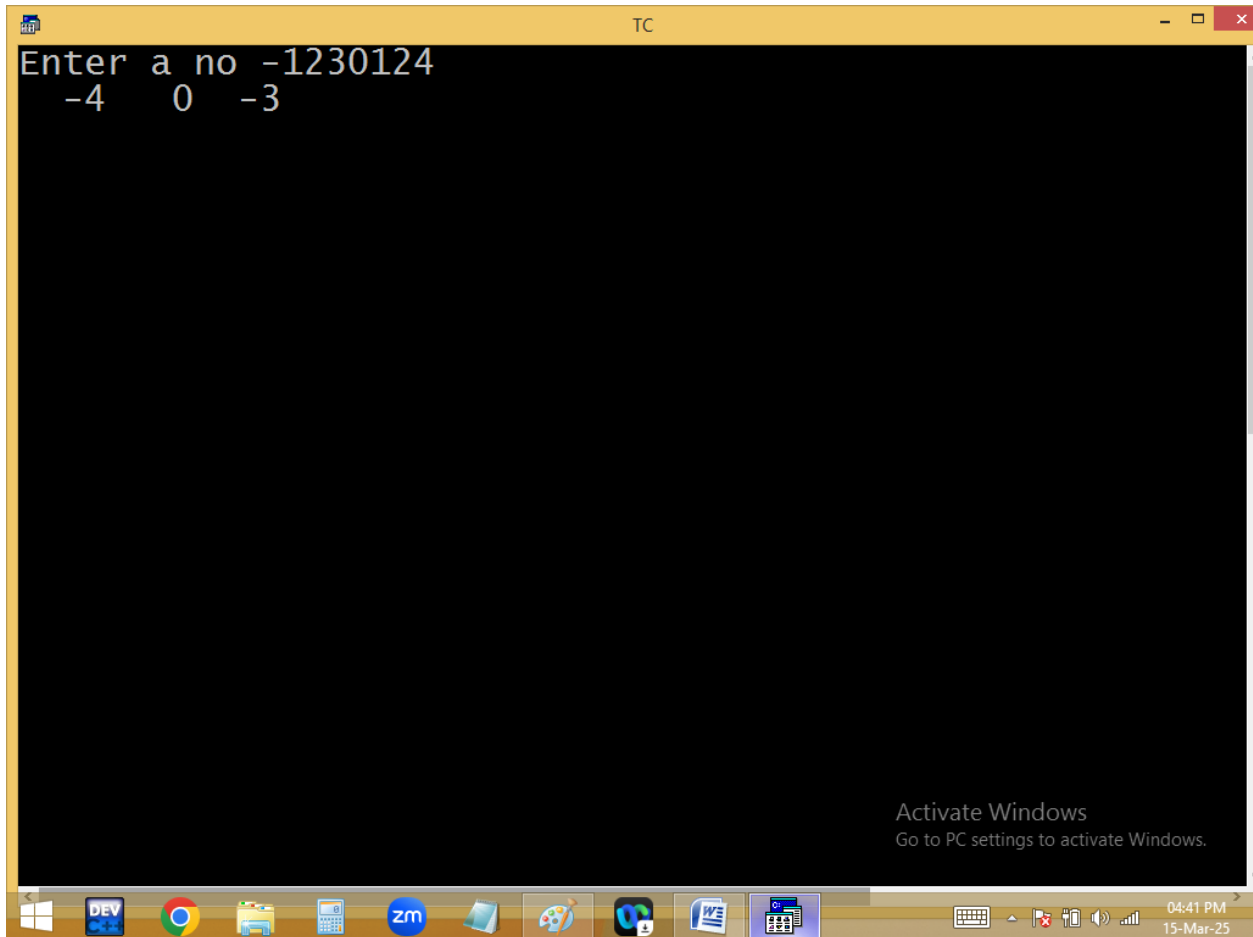
	<u>i</u>	<u>n</u>	<u>m</u>	<u>c</u>
	0	123125	123125	012
for(i=0; i<=9; i++)	1			012
{	2			01
for(c=0, m=n; m!=0;m=m/10)	3			01
{	5			
if(i==m%10)c++;				
}				
if(c==1) p(i);				
}				

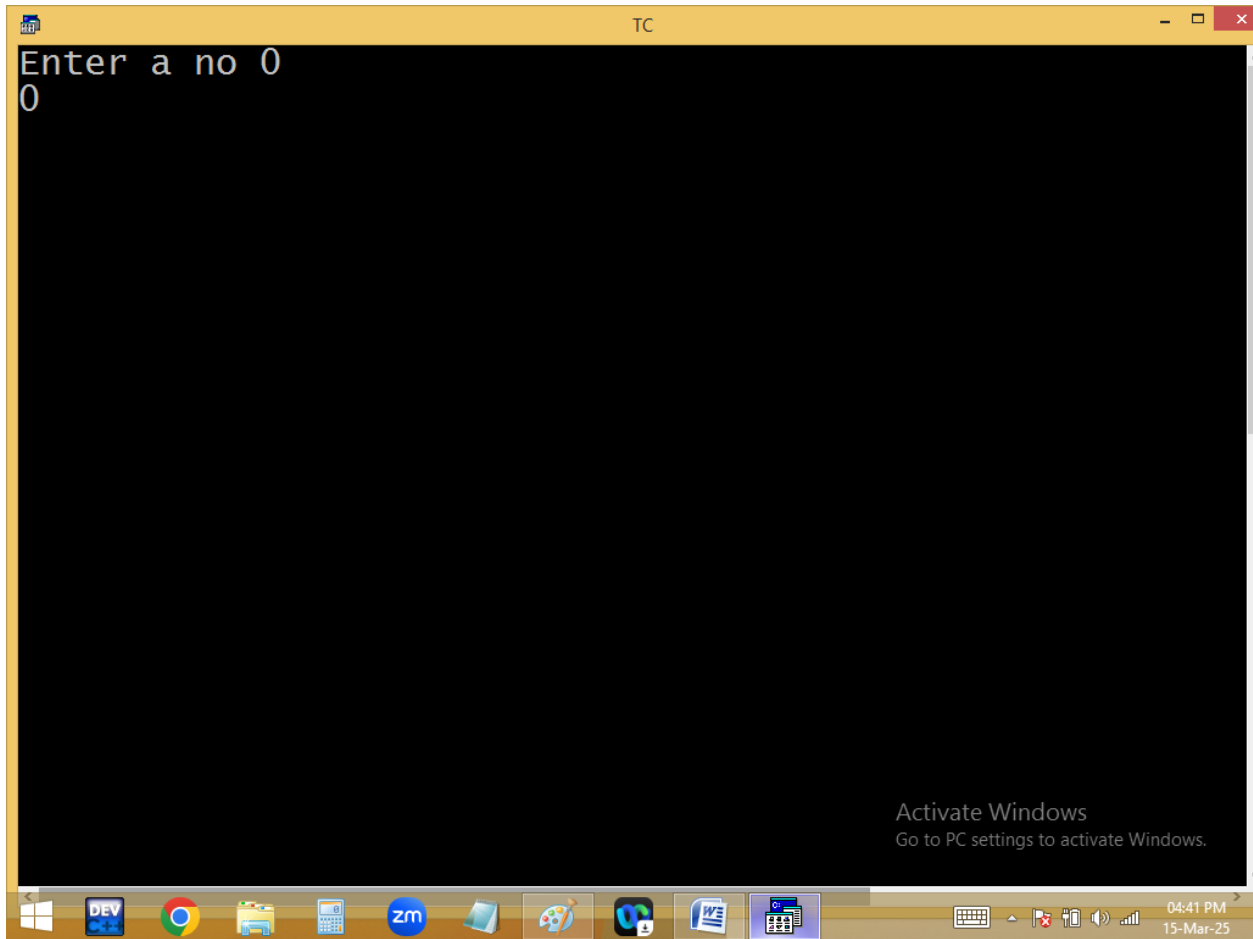
```
TC
File Edit Run Compile Project Options Deb
Line 5 Col 15 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,m,i,c,a;
    clrscr();
    printf("Enter a no "); scanf("%ld",&n);
    if(n==0) printf("0");
    for(m=n;m!=0;m=m/10)
    {
        for(c=0,a=n;a!=0;a=a/10)
        {
            if(a%10==m%10) c++;
        }
        if( c==1 )printf("%4d",m%10);
    }
    getch();
}
```

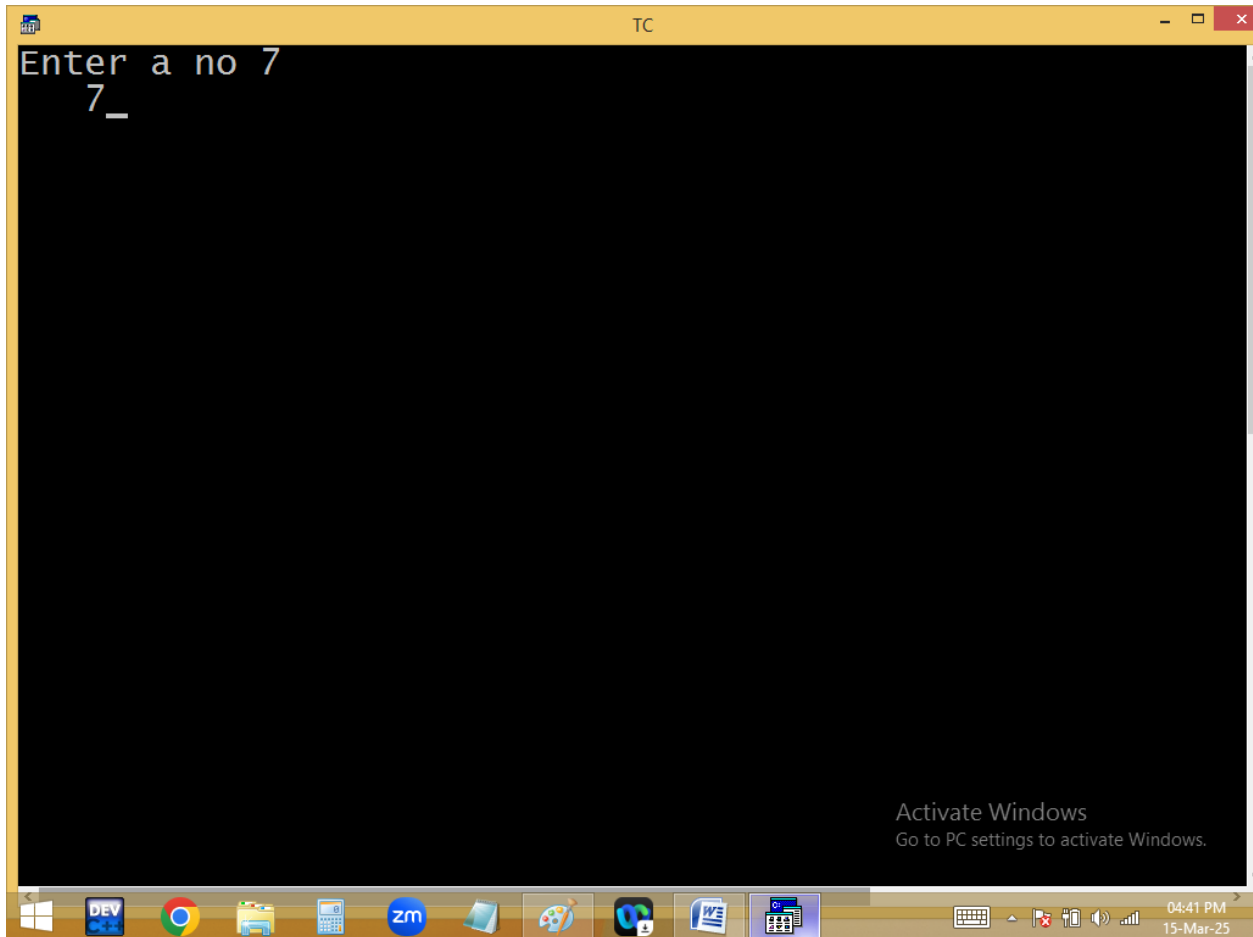
Activate Windows
Go to PC settings to activate Windows.

04:40 PM
15-Mar-25









```

for( m=n; m!=0; m/=10 )
{
for( c=0, a=n; a!=0; a/=10)
{
if( a%10==m%10)c++;
}
if(c==1) printf("%d",m%10);
}

```

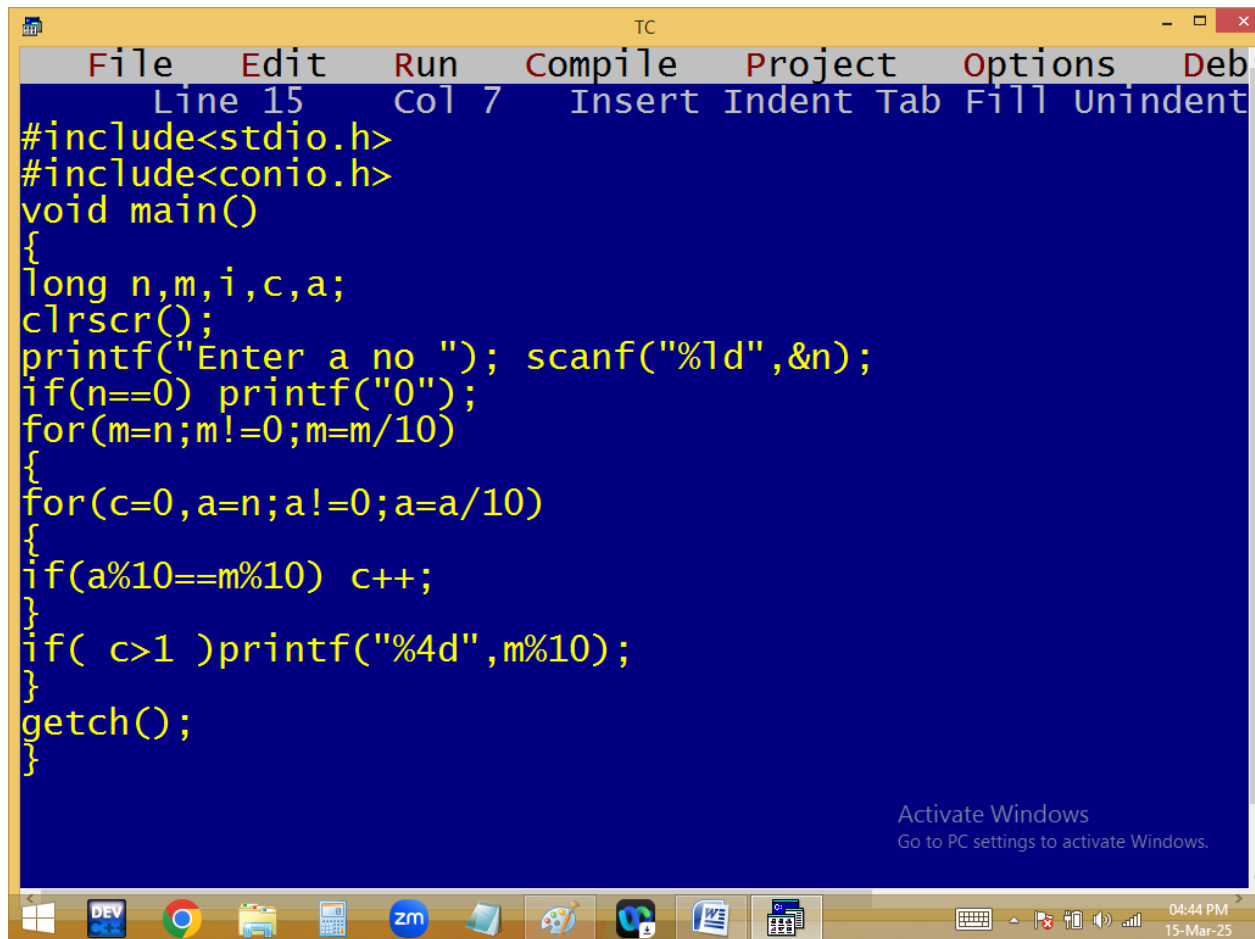
Handwritten annotations above the code:

- m above `m=n`
- m above `m!=0`
- a above `a=n`
- c above `c=0`

Handwritten annotations to the right of the code:

- 123125 under `m=n`
- 123125 under `m!=0`
- 123125 under `a=n`
- 0 1 under `c=0`
- 0 under `a=n`

Duplicate no's:



```
TC
File Edit Run Compile Project Options Deb
Line 15 Col 7 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
long n,m,i,c,a;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
if(n==0) printf("0");
for(m=n;m!=0;m=m/10)
{
for(c=0,a=n;a!=0;a=a/10)
{
if(a%10==m%10) c++;
}
if( c>1 )printf("%4d",m%10);
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

04:44 PM
15-Mar-25

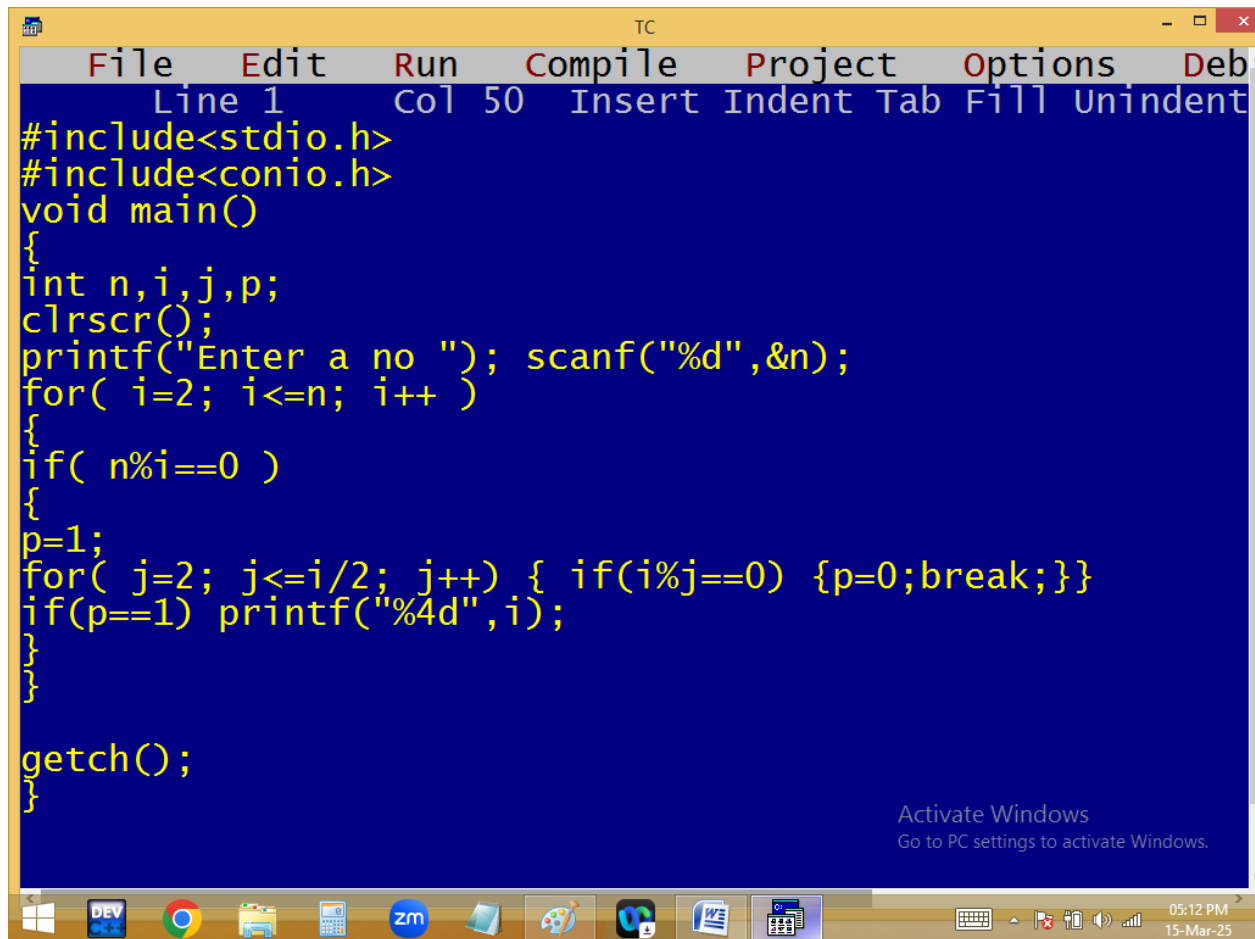
The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text displays the input number and its prime factors: "Enter a no 111223" followed by "2 2 1 1 1" on the next line. The window title bar says "TC". The Windows taskbar at the bottom shows various icons including DEV, Chrome, File Explorer, Calculator, Zoom, and others. The system clock in the bottom right corner indicates 04:44 PM on 15-Mar-25. An "Activate Windows" watermark is visible in the bottom right of the TC window.

```
Enter a no 111223
2 2 1 1 1
```

Finding prime factors of given no?

8 factors are 1, 2, 4, 8

10 factors are 1,2,5,10

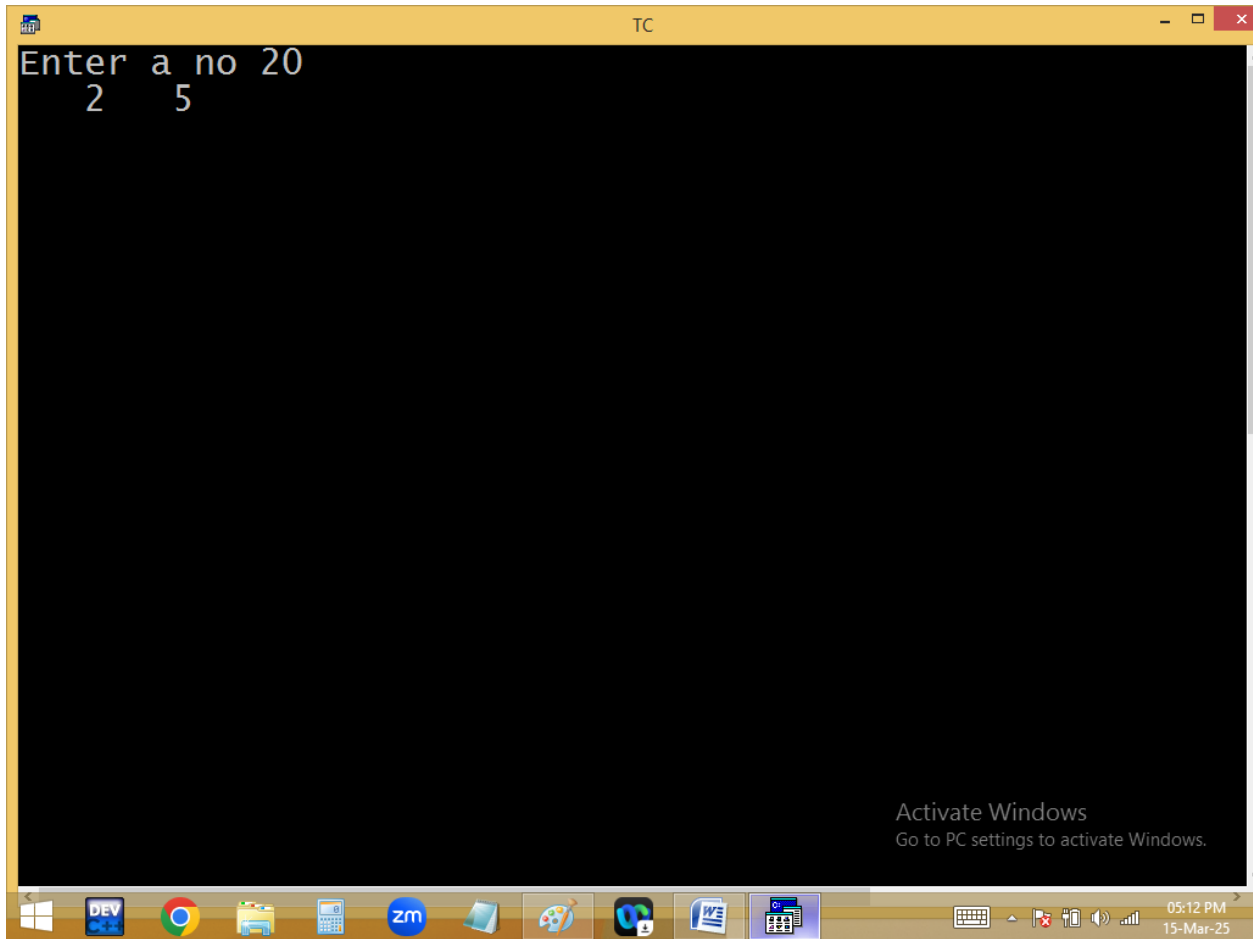


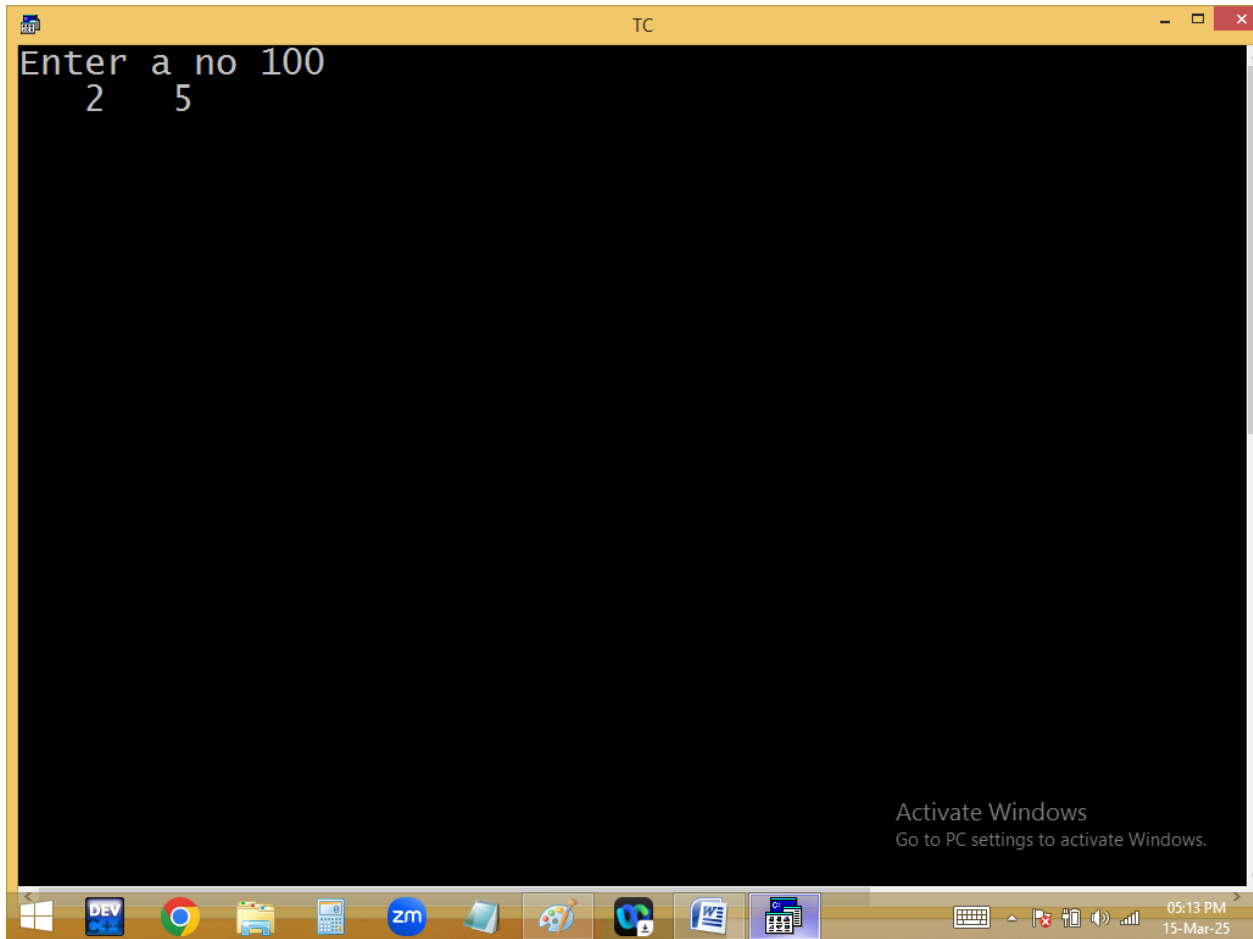
```
TC
File Edit Run Compile Project Options Deb
Line 1 Col 50 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,p;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for( i=2; i<=n; i++ )
{
if( n%i==0 )
{
p=1;
for( j=2; j<=i/2; j++) { if(i%j==0) {p=0;break;}}
if(p==1) printf("%4d",i);
}
}

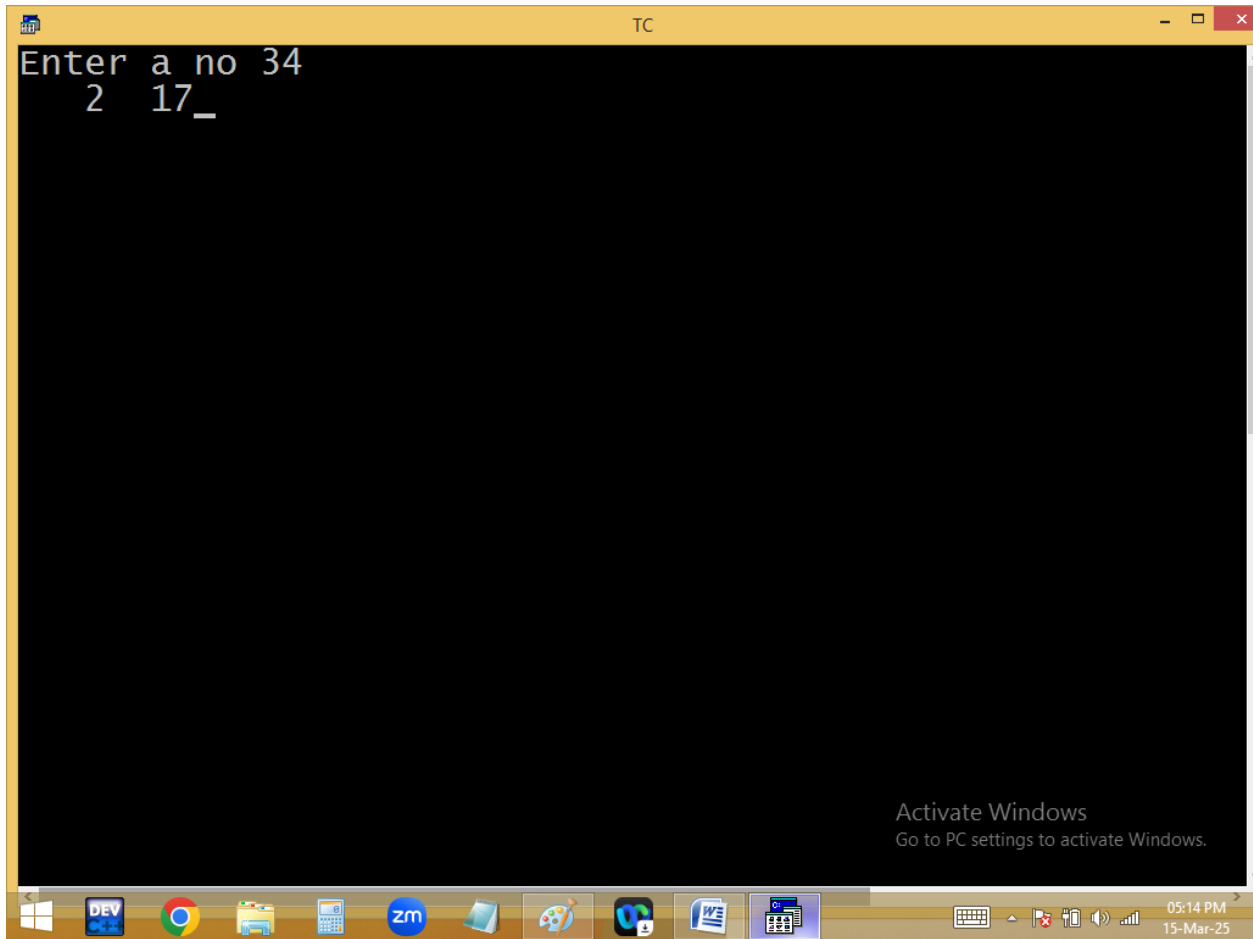
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

05:12 PM
15-Mar-25







for(i=2; i<=n; i++)

{

if(n%i==0) ✓

{

p=1; ✓✓

for(j=2; j<=i/2; j++) {if(i%j==0)p=0;break;}

if(p==1) p(i);

}

}

$$\begin{array}{l} \frac{n}{i} \quad \frac{i}{j} \quad \frac{p}{j} \quad \frac{j}{2} \leq \frac{2}{2} = 1 \\ 20 \div 2 = 0 \quad \frac{2}{2} \\ 20 \div 4 = 0 \quad \frac{2}{2} \\ 20 \div 5 = 0 \\ 20 \div 10 = 0 \end{array}$$

$$\begin{array}{l} \frac{i}{j} \quad \frac{p}{j} \\ 10 \div 1 = 0 \quad \frac{p}{1} \end{array}$$

2 ~~4~~ 5 10

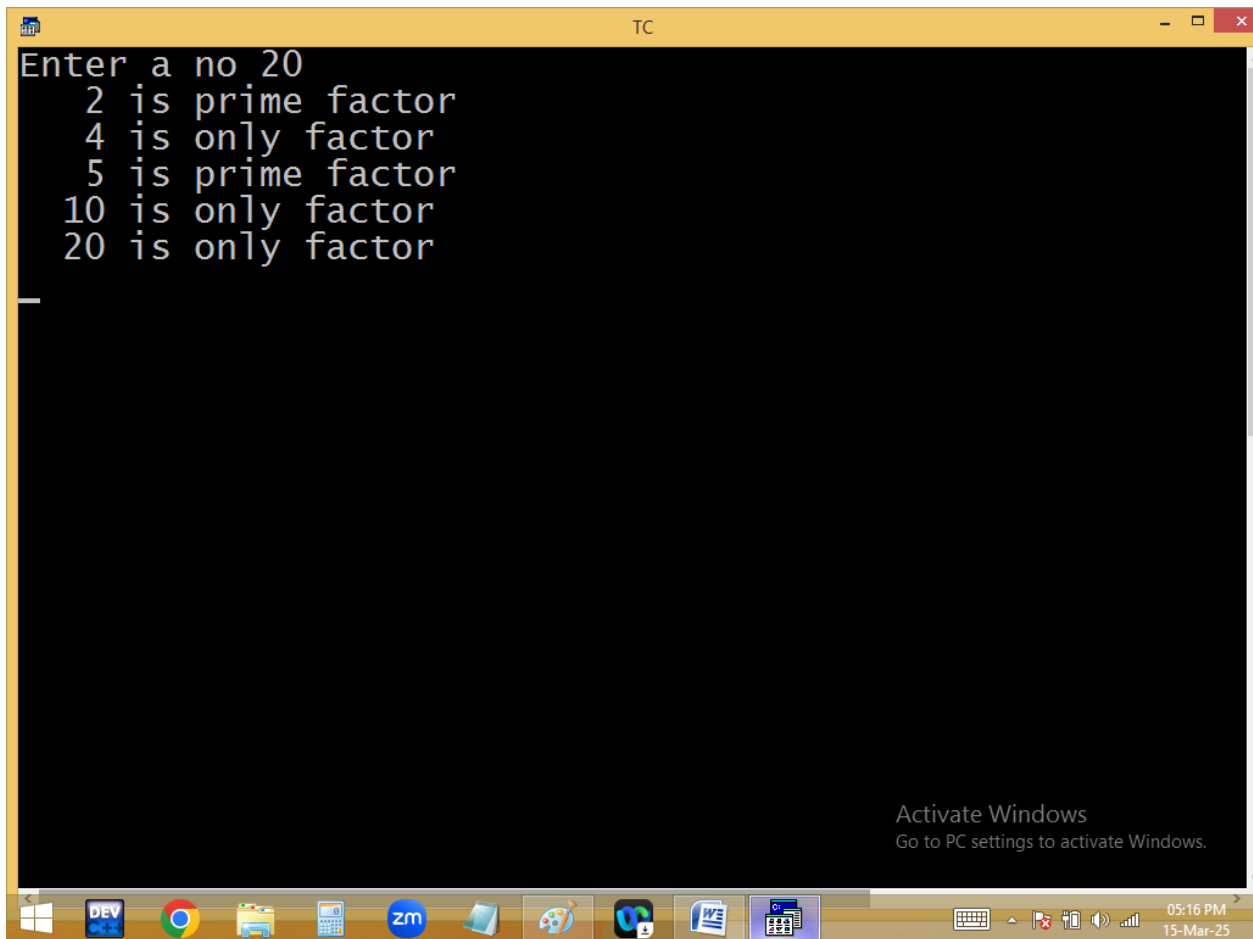
```

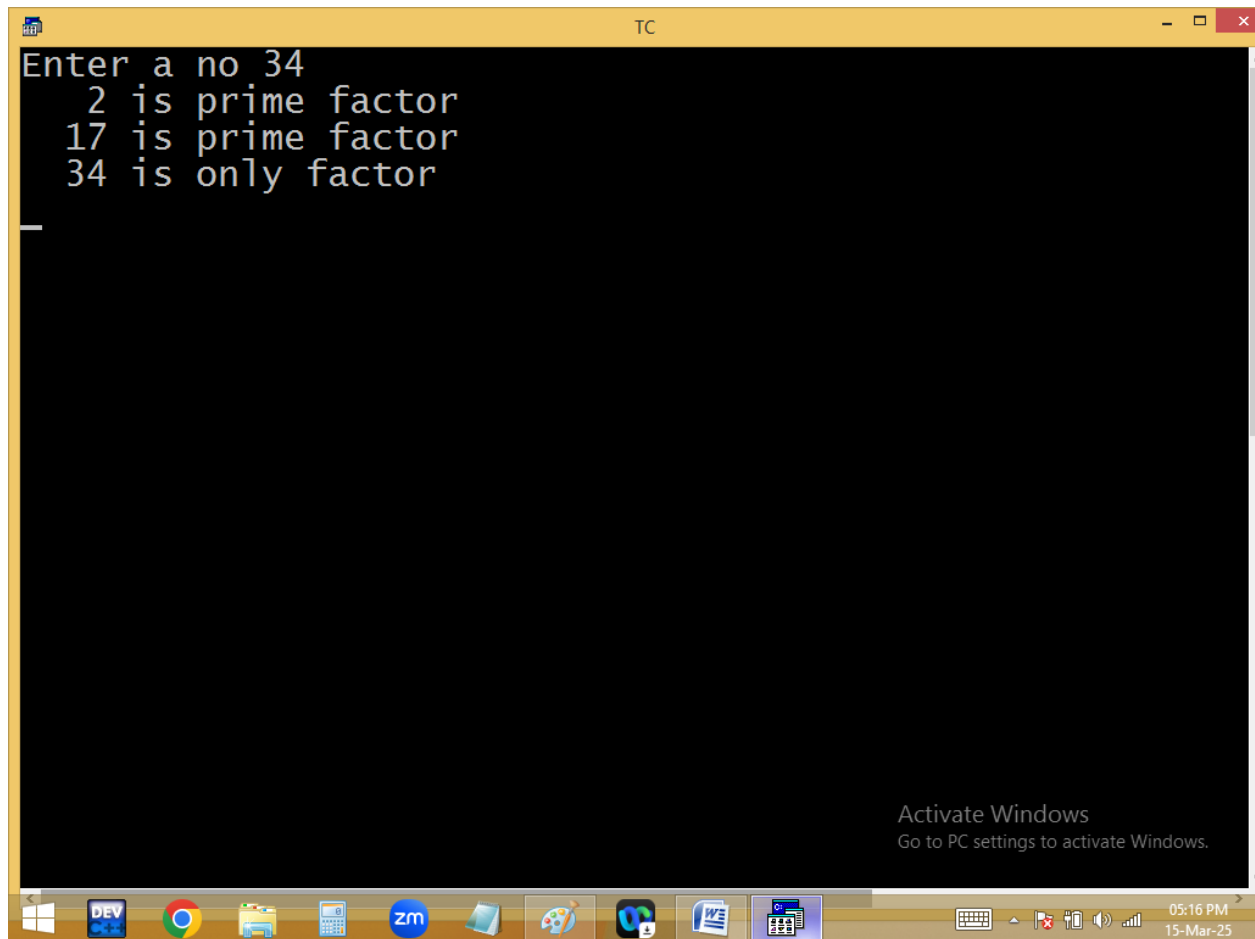
TC
File Edit Run Compile Project Options Deb
Line 18 Col 1 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,j,p;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for( i=2; i<=n; i++ )
{
if( n%i==0 )
{
p=1;
for( j=2; j<=i/2; j++) { if(i%j==0) {p=0;break;}}
if(p==1) printf("%4d is prime factor\n",i);
else printf("%4d is only factor\n",i);
}
}
getch();
}

```

Activate Windows
Go to PC settings to activate Windows.

05:15 PM
15-Mar-25

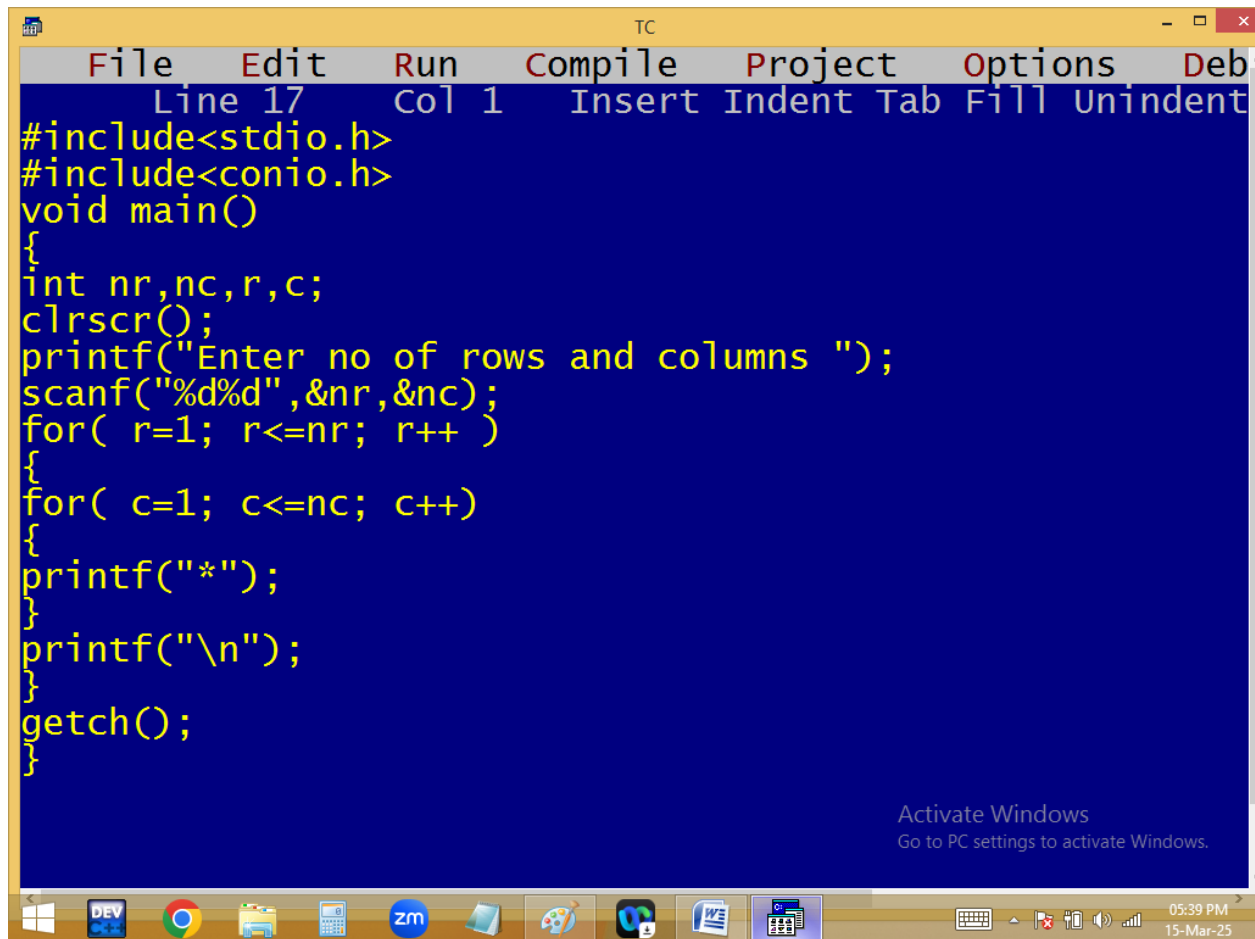




A screenshot of a Windows 10 desktop. A terminal window titled 'TC' is open, displaying the following text:
Enter a no 34
2 is prime factor
17 is prime factor
34 is only factor
Below the terminal window, the Windows taskbar is visible with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and several other applications. The system tray on the right shows the date and time as 05:16 PM on 15-Mar-25. An 'Activate Windows' watermark is visible in the bottom right corner of the terminal window.

```
Enter a no 34
2 is prime factor
17 is prime factor
34 is only factor
```

Patterns:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 17", "Col 1", and a list of actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is a C program that prompts the user for the number of rows and columns, then prints a grid of asterisks. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        for( c=1; c<=nc; c++)
        {
            printf("*");
        }
        printf("\n");
    }
    getch();
}
```

At the bottom right of the editing area, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 05:39 PM on 15-Mar-25.

```

Enter no of rows and columns 3 4
****
****
****

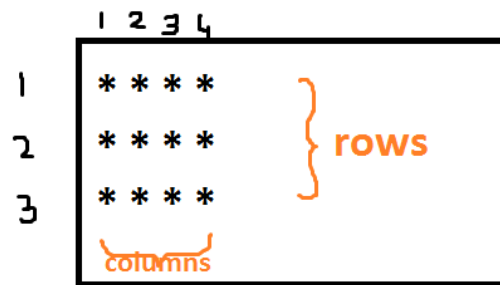
```

Activate Windows
Go to PC settings to activate Windows.

```

for( r=1; r<=nr ; r++ )
{
for( c=1; c<=nc; c++ )
{
printf( " * " );
}
p("\n");
}

```



$$\frac{nr}{3} \quad \frac{nc}{4}$$

$$\frac{r=1 \leq nr}{1 \leq 3}$$

$$\frac{c=1 \leq nc}{1 \ 2 \ 3 \ 4}$$

$$2 \leq 3$$

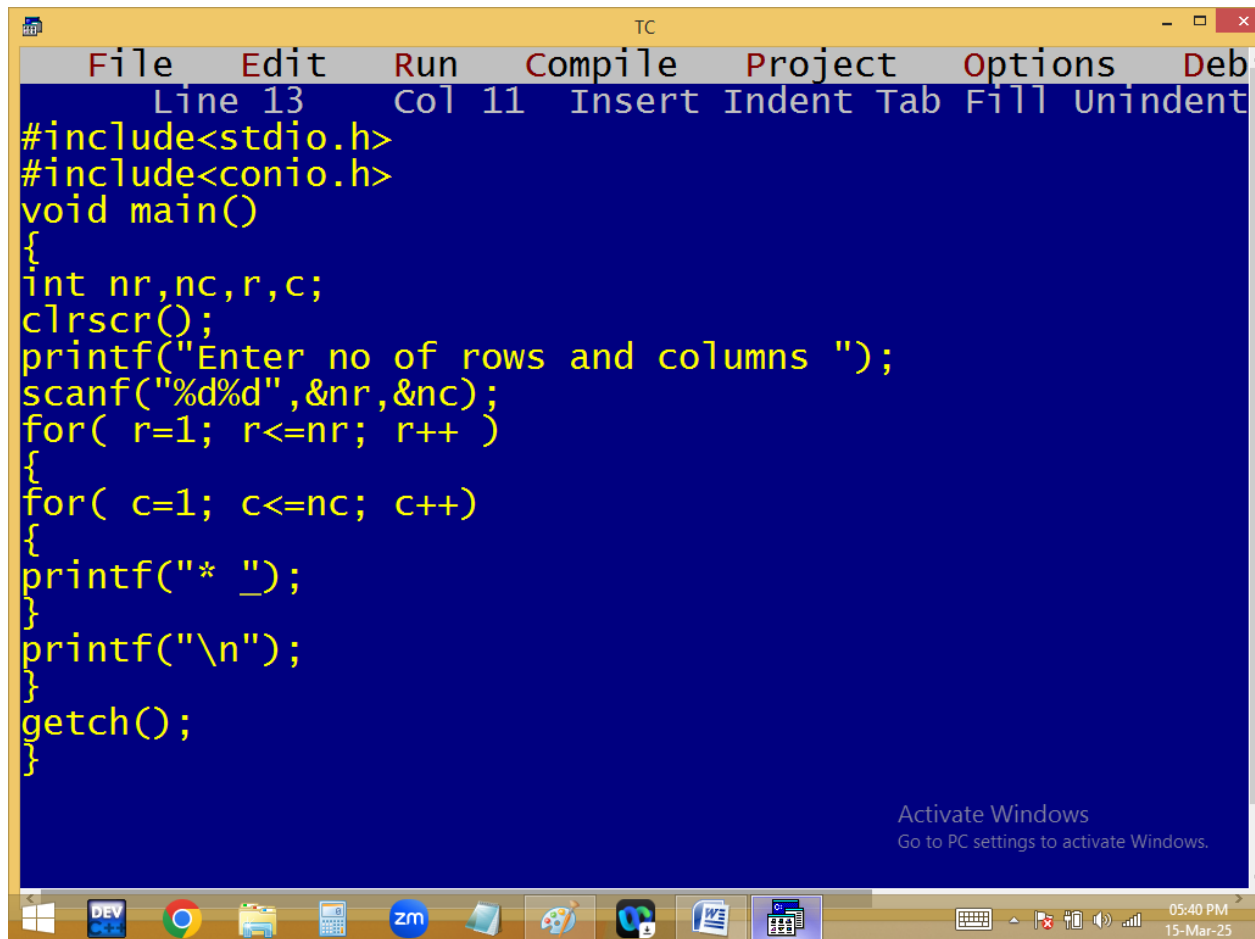
$$\begin{matrix} * * * * \\ 1 \ 2 \ 3 \ 4 \end{matrix}$$

$$3 \leq 3$$

$$\begin{matrix} * * * * \\ 1 \ 2 \ 3 \ 4 \end{matrix}$$

$$4 \leq 3$$

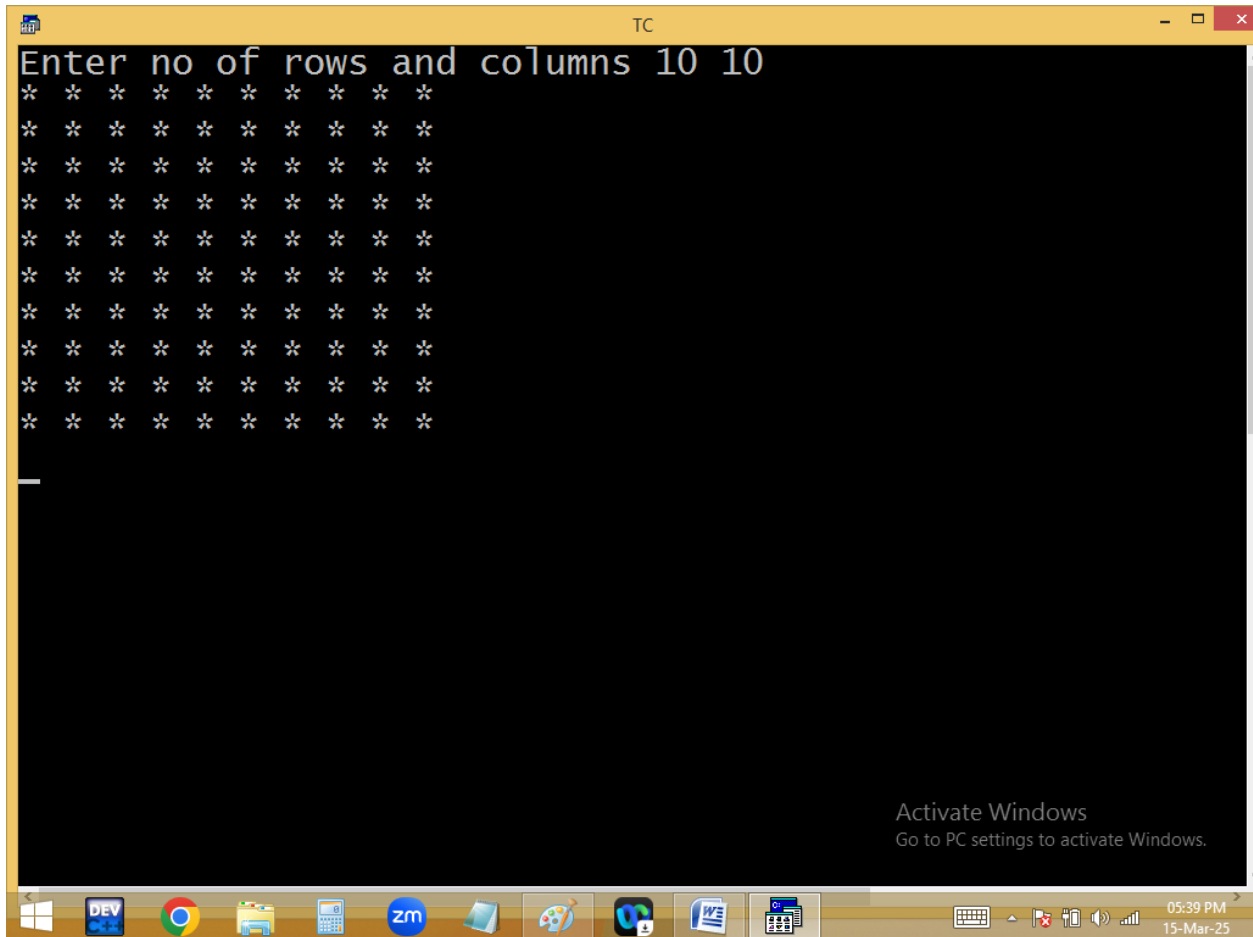
$$\begin{matrix} * * * * \\ 1 \ 2 \ 3 \ 4 \end{matrix}$$

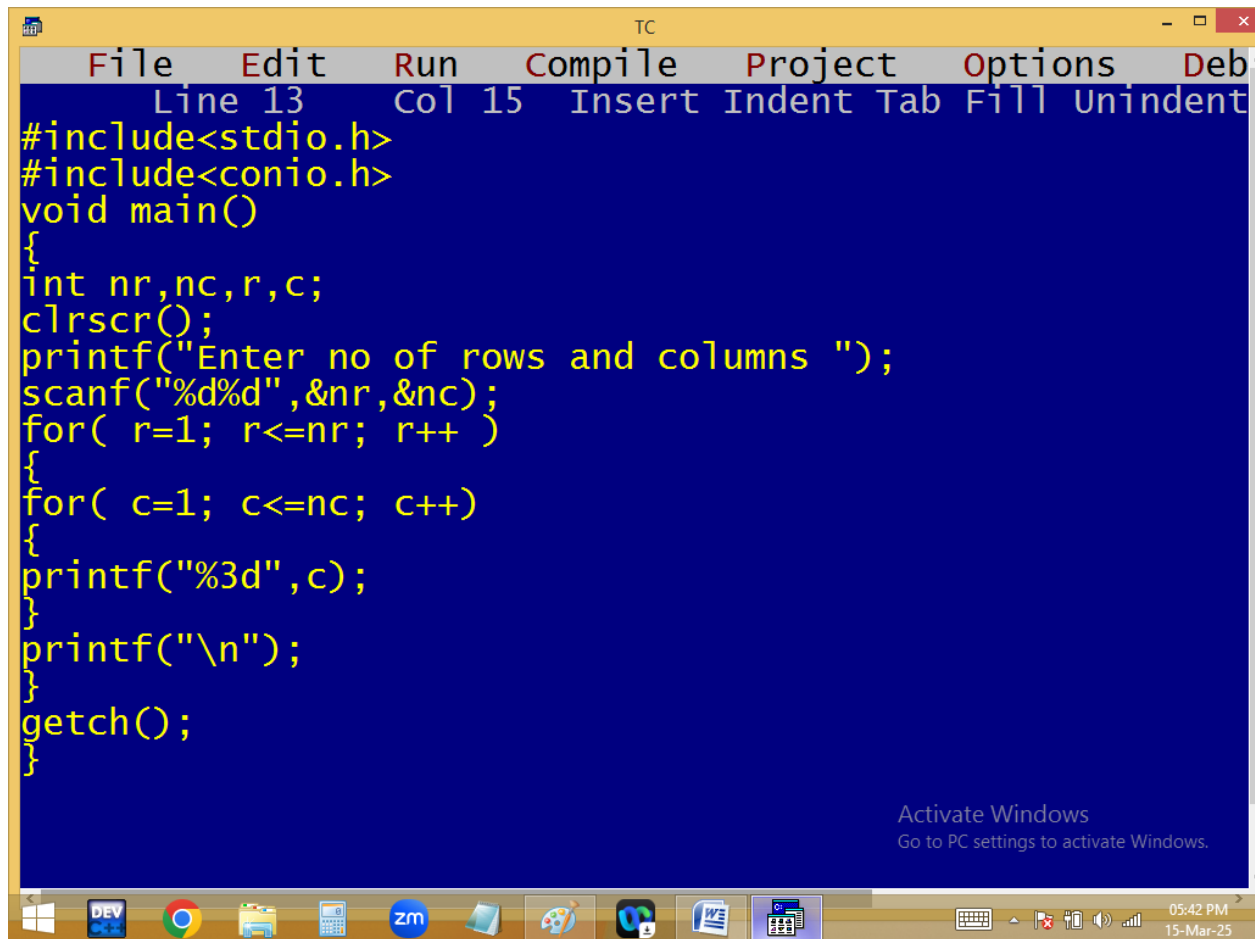


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar displays "Line 13", "Col 11", and a list of actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is a C program that prompts the user for the number of rows and columns, then prints a grid of asterisks. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        for( c=1; c<=nc; c++)
        {
            printf("* ");
        }
        printf("\n");
    }
    getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 05:40 PM on 15-Mar-25.

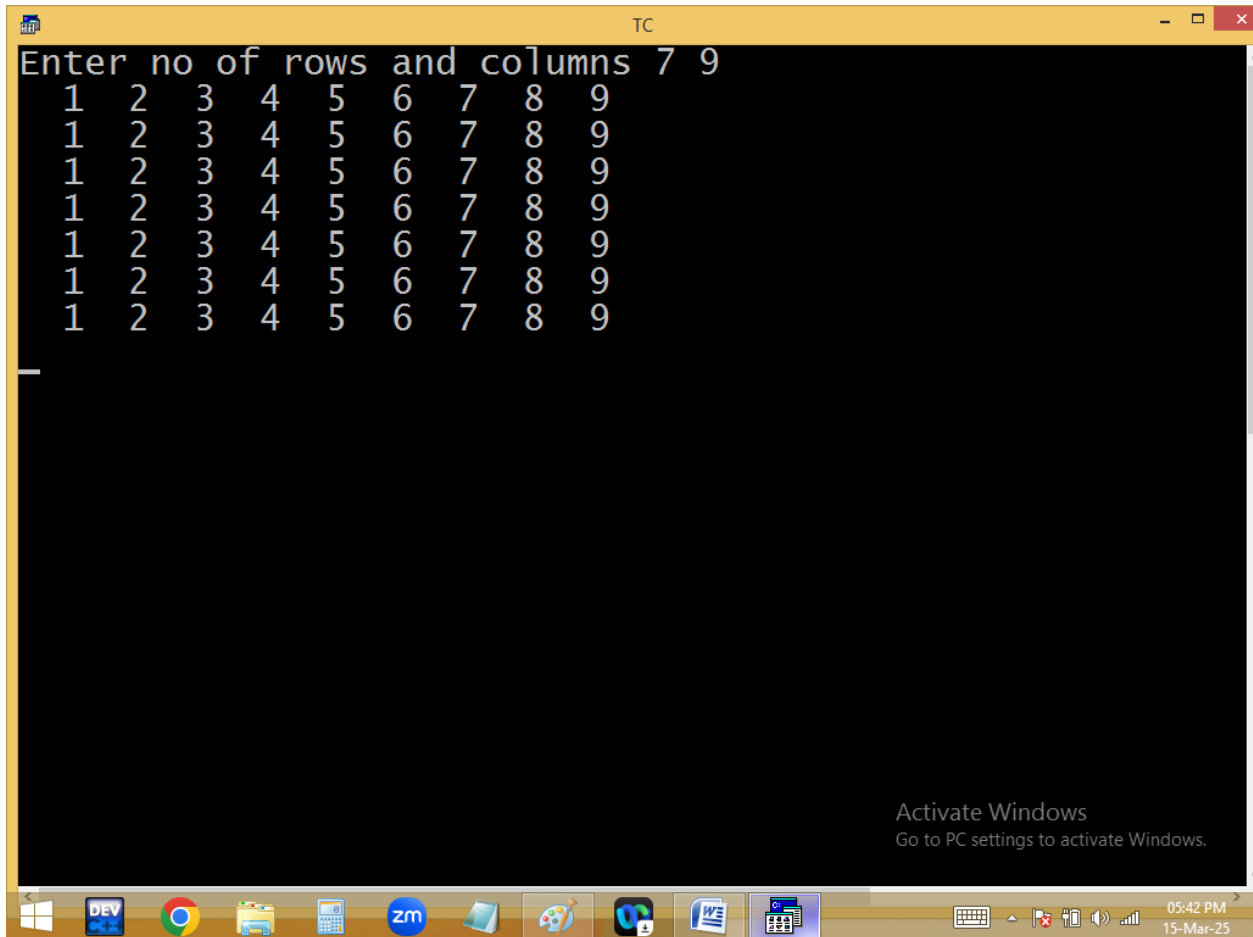


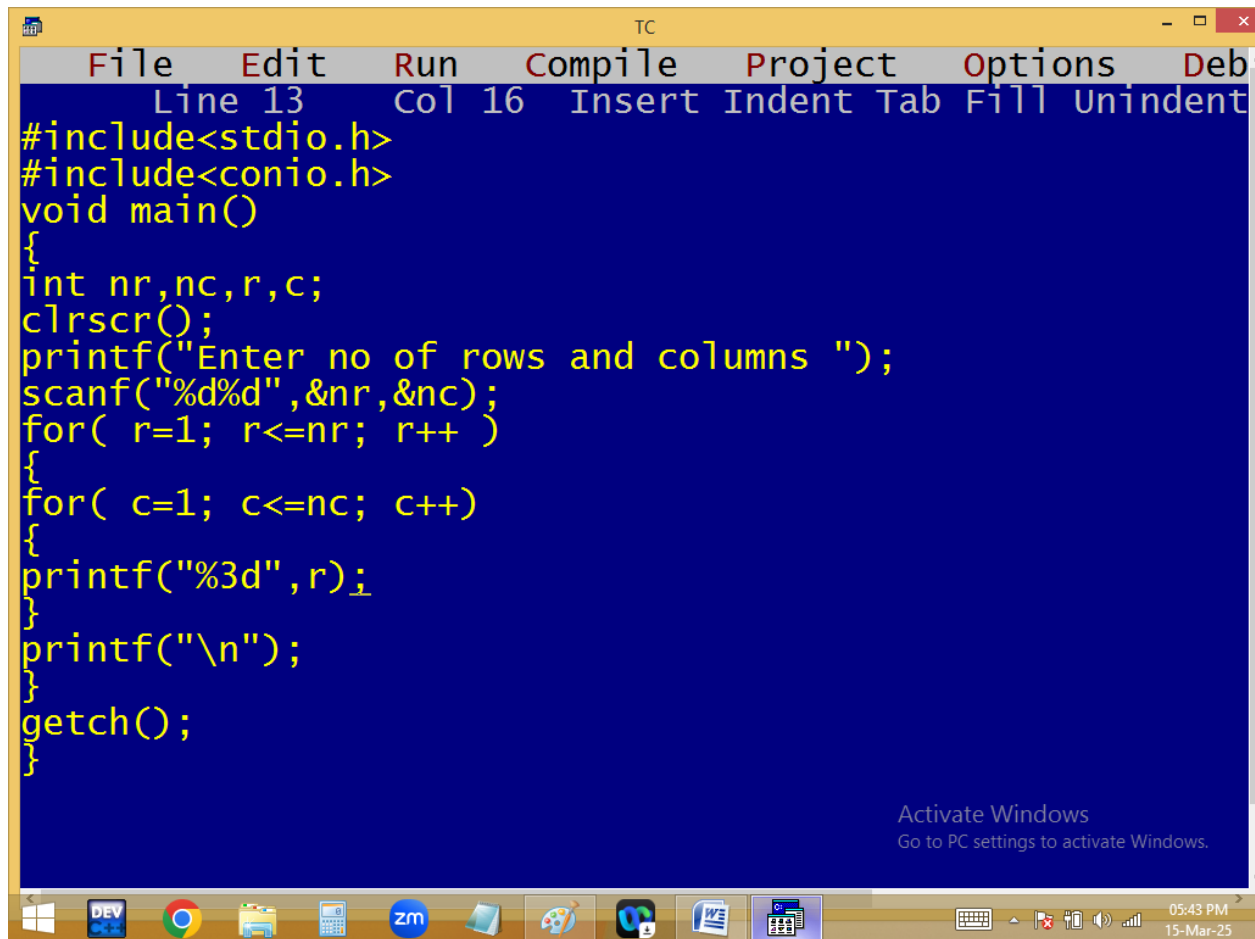


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 15", and a list of actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of numbers from 1 to the total number of elements. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        for( c=1; c<=nc; c++)
        {
            printf("%3d",c);
        }
        printf("\n");
    }
    getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 05:42 PM on 15-Mar-25.

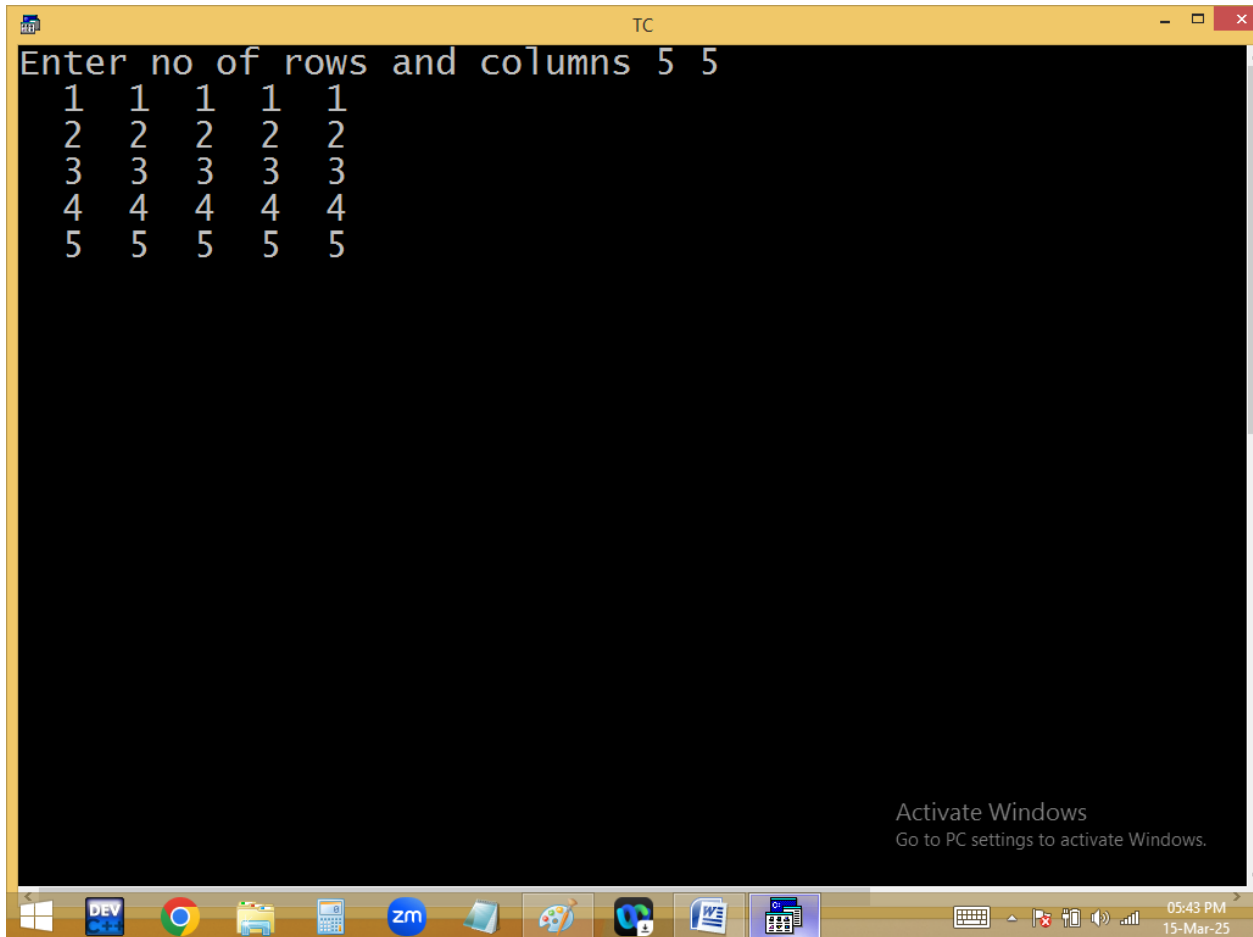


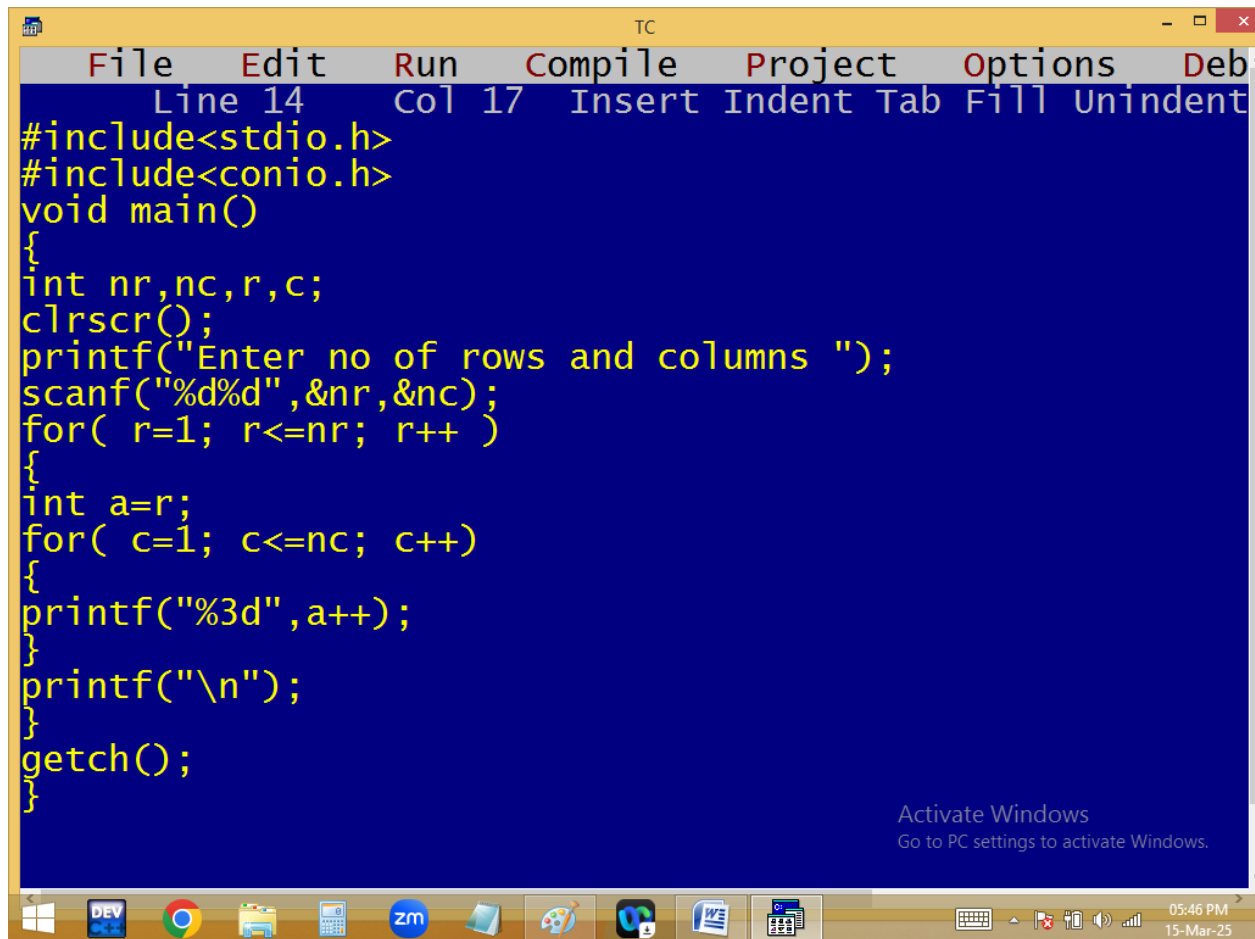


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 13", "Col 16", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. The code is a C program that takes the number of rows and columns as input and prints a grid of numbers from 1 to the total number of elements. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        for( c=1; c<=nc; c++)
        {
            printf("%3d",r);
        }
        printf("\n");
    }
    getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 05:43 PM on 15-Mar-25.

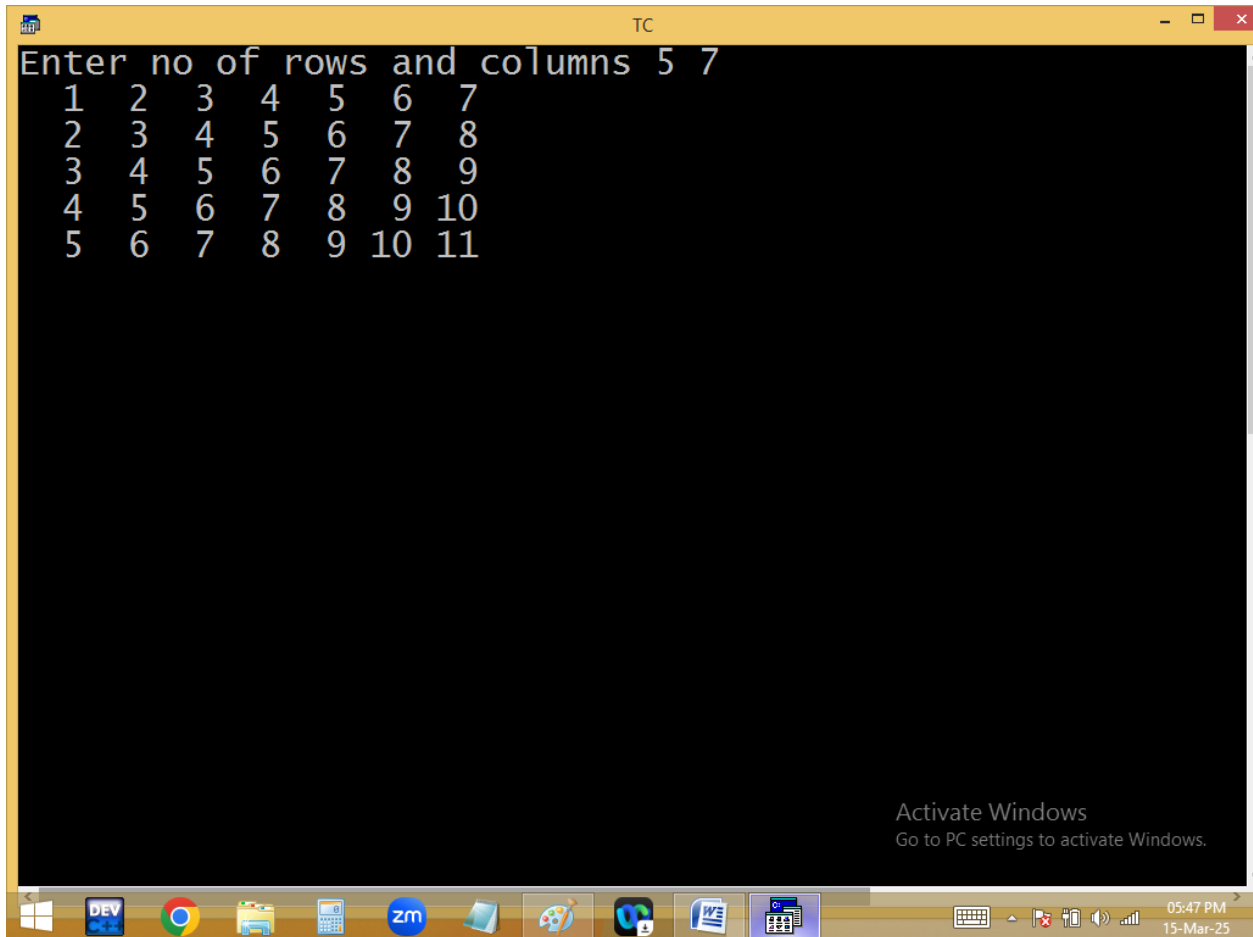


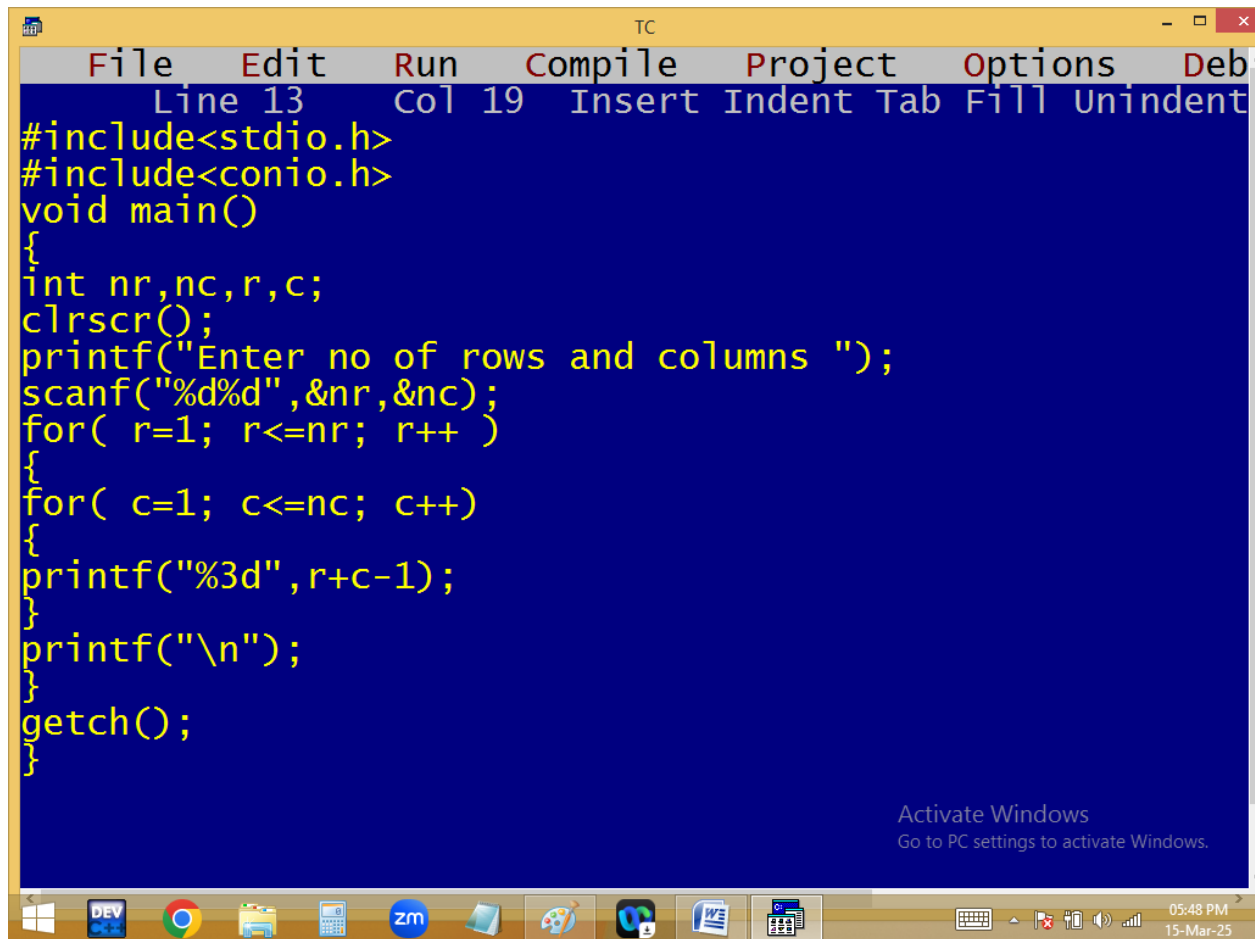


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 14", "Col 17", and provides options: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        int a=r;
        for( c=1; c<=nc; c++)
        {
            printf("%3d",a++);
        }
        printf("\n");
    }
    getch();
}
```

In the bottom right corner of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "05:46 PM" and "15-Mar-25".





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 13", "Col 19", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", and "Unindent". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr,nc,r,c;
    clrscr();
    printf("Enter no of rows and columns ");
    scanf("%d%d",&nr,&nc);
    for( r=1; r<=nr; r++ )
    {
        for( c=1; c<=nc; c++)
        {
            printf("%3d",r+c-1);
        }
        printf("\n");
    }
    getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the very bottom, showing icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner shows "05:48 PM" and "15-Mar-25".

```
TC
Enter no of rows and columns 10 10
1 2 3 4 5 6 7 8 9 10
2 3 4 5 6 7 8 9 10 11
3 4 5 6 7 8 9 10 11 12
4 5 6 7 8 9 10 11 12 13
5 6 7 8 9 10 11 12 13 14
6 7 8 9 10 11 12 13 14 15
7 8 9 10 11 12 13 14 15 16
8 9 10 11 12 13 14 15 16 17
9 10 11 12 13 14 15 16 17 18
10 11 12 13 14 15 16 17 18 19
```

Activate Windows
Go to PC settings to activate Windows.

Home work:

1 2 3 4

4 3 2 1

1 2 3 4

4 3 2 1

Eg2:

ABCD

ABCD

ABCD

Eg3

ABCD

abcd

ABCD

abcd

Eg4:

AbCdEf

AbCdEf

AbCdEf

